A UNIFORM CLOSED FORM FOR MODULAR SCHUR NUMBERS $S_m(k,\ell)$

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Abstract. The modular Schur number $S_m(k,\ell)$ of Chappelon–Marchena–Domínguez [2] is the greatest N such that $[1,N]$ admits a k-partition into classes that are ℓ-sum-free modulo m. We prove that

$$S_m(k,\ell) \;=\; \frac{m}{\gcd(m,\ell-1)} - 1$$

for every $m \geq 2$, $\ell \geq 2$, and $k \geq m/\gcd(m,\ell-1) - 1$. The proof is two paragraphs of modular arithmetic. It subsumes the per-modulus tables computed in [2] ($m \in \{1,2,3\}$) and [3] ($m \in \{4,5,6,7\}$), and extends them to all m. The main theorem is formalized in Lean 4 with axiom dependencies only `propext`, `Classical.choice`, `Quot.sound`. We also give a sharp k_0 for prime moduli and record a small composite counterexample ($C = \{1,5\} \subseteq \mathbb{Z}/\mathrm{m}[12]$) that refutes the natural "only singletons are safe" conjecture.

1. Introduction

The classical Schur number $S(k)$ is the largest N such that $\{1,\ldots,N\}$ admits a k-partition into sum-free sets. Values are known only through $S(5) = 160\,537$ [4]; $S(6)$ is the content of Erdős problem 483 and remains open. Generalizations to longer equations were studied by Beutelspacher and Brestovansky [1] and are now textbook material [5].

Chappelon, Marchena, and Domínguez [2] introduced the modular version of the generalized Schur number.

Definition 1.1. A set $S \subseteq \mathbb{Z}$ is *ℓ-sum-free modulo m* if no $x_1,\ldots,x_\ell \in S$ (repetitions allowed) and $y \in S$ satisfy $x_1 + \cdots + x_\ell \equiv y \pmod{m}$. The *modular Schur number* $S_m(k,\ell)$ is the greatest $N \geq 0$ such that $[1,N]$ admits a k-partition into ℓ-sum-free sets modulo m.

Chappelon et al. [2] closed $m \in \{1,2,3\}$ and observed that the determination of $S_m(k,\ell)$ "seems to be much more difficult for moduli $m \geq 4$" ([2, §6]). D'orville, Sim, Wong, and Ho [3] subsequently closed $m \in \{4,5,6,7\}$ via residue-by-residue case analysis on $\ell \bmod m$, leaving $m \geq 8$ as their open problem 1.5. The pattern visible across all ten tables—once k is large enough, $S_m(k,\ell)$ is a function of $\gcd(m,\ell-1)$ alone—was not formulated as a unified statement.

Writing $d := \gcd(m,\ell-1)$ and $n := m/d$, the following theorem makes that pattern explicit, with no hypothesis on m and no case split on $\ell \bmod m$.

Theorem 1.2. *For every $m \geq 2$, $\ell \geq 2$, and $k \geq n-1$,*

$$S_m(k,\ell) \;=\; n-1 \;=\; \frac{m}{\gcd(m,\ell-1)} - 1.$$

The proof is elementary: both bounds reduce to the observation that n itself is a singleton that fails ℓ-sum-freeness. We give it in §3–§4.

The formula looks natural in retrospect, but the prior case-by-case framework obscures it: the proofs in [2, 3] fix a residue class $\ell \bmod m$ and chase a witness partition residue by residue, which hides the single observation that "the element $n = m/\gcd(m,\ell-1)$ is unsafe as a singleton." The argument in §3 is one paragraph because it stops at that observation. The actual combinatorial content is the threshold k_0: §5 gives it tightly for prime m and shows (via the explicit witness $\{1,5\} \subseteq \mathbb{Z}/\mathrm{m}[12]$) that the prime case does not generalize naively.

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Formalization. The proof is short but has a form that is easy to misstate (an earlier draft confused the role of n versus m in the forbidden tuple). I formalized Theorem 1.2 in Lean 4 to remove any doubt; the proof was completed by Aristotle [6]. The Lean statement

```
theorem schurModResidue_eq (m k l : N) (hm : 2 <= m) (hl : 2 <= l)
    (hk : m / Nat.gcd m (l - 1) - 1 <= k) :
    schurModResidue m k l = m / Nat.gcd m (l - 1) - 1
```

compiles with axiom dependencies only `propext`, `Classical.choice`, `Quot.sound`. Source is at `https://github.com/mysticflounder/modular-schur`.

Organization. §2 collects base lemmas. §3 and §4 prove Theorem 1.2. §5 handles the threshold $k_0(m, \ell)$, including the sharp prime-modulus case and a composite counterexample. §6 specializes to $m \in \{8, 9, 11, p^e\}$. §7 lists what stays open.

2. Preliminaries

Throughout, $m \geq 2$ and $\ell \geq 2$ are fixed integers, $d := \gcd(m, \ell - 1)$, and $n := m/d$. Note $dn = m$ exactly, and $d \mid (\ell - 1)$ by definition.

Lemma 2.1 (Residue reduction)**.** *If $N \leq m - 1$, the integers $1, 2, \ldots, N$ have pairwise distinct nonzero residues modulo m. A partition of $[1, N]$ into classes $C_1 \sqcup \cdots \sqcup C_k$ is valid if and only if each C_i, viewed as a subset of $\{1, \ldots, m - 1\} \subseteq \mathbb{Z}/\mathrm{m}$, is ℓ-sum-free modulo m.*

Proof. All elements of $[1, N]$ lie in $[1, m - 1]$, and the map $[1, m - 1] \to \mathbb{Z}/\mathrm{m}$ is injective onto $\{1, \ldots, m - 1\}$. Sum-freeness is preserved under this identification. □

Lemma 2.2 (Universal upper bound, [2])**.** *For every $\ell \geq 1$ and every $k \geq 1$, $S_m(k, \ell) \leq m - 1$.*

Proof. If $N \geq m$, then $m \in [1, N]$ lies in some class C. The tuple $(m, \ldots, m; m) \in C^{\ell+1}$ satisfies $m \cdot \ell \equiv 0 \equiv m \pmod{m}$, violating ℓ-sum-freeness. (This is [2, Eq. (2)].) □

Lemma 2.3 (Singleton safety)**.** *For $r \in \{1, \ldots, m - 1\}$, the singleton $\{r\} \subseteq \mathbb{Z}/\mathrm{m}$ is ℓ-sum-free modulo m if and only if $(\ell - 1)r \not\equiv 0 \pmod{m}$; equivalently, if and only if $n \nmid r$.*

Proof. $\ell\{r\} \cap \{r\} \neq \emptyset$ iff $\ell r \equiv r \pmod{m}$ iff $(\ell - 1)r \equiv 0 \pmod{m}$. Write $\ell - 1 = du$ so $\gcd(u, n) = 1$; then $(\ell - 1)r \equiv 0 \pmod{dn}$ iff $ur \equiv 0 \pmod{n}$ iff $n \mid r$. □

Lemma 2.4 (Monotonicity in k)**.** *If $[1, N]$ admits a valid k_0-partition, then for every $k \geq k_0$ it admits a valid k-partition.*

Proof. Pad with $k - k_0$ empty classes; the empty set is vacuously ℓ-sum-free. □

3. Upper bound

Theorem 3.1. *For all $m \geq 2$, $\ell \geq 2$, and $k \geq 1$,*

$$S_m(k, \ell) \;\leq\; n - 1.$$

Proof. Suppose $N \geq n$. Since $1 \leq n \leq m$, we have $n \in [1, N]$; let C be the class of a valid k-partition containing n. Write $\ell - 1 = du$ with $u \in \mathbb{Z}$, so

$$(\ell - 1) \cdot n \;=\; du \cdot n \;=\; u \cdot (dn) \;=\; u \cdot m \;\equiv\; 0 \pmod{m}.$$

Hence $\ell n \equiv n \pmod{m}$. The tuple $(n, \ldots, n; n) \in C^{\ell+1}$ thus witnesses ℓ-sum-freeness failure in C, a contradiction. □

Remark 3.2. The two obstructions in prior arguments—"$N \geq m$ is blocked by m" (Lemma 2.2, the $d = 1$ case $n = m$) and "$r = m/d$ is an unsafe singleton" (Lemma 2.3 at $r = n$)—are the same obstruction. Both reduce to: the single element n is unsafe as a singleton, so any class containing it is unsafe. Once $N \geq n$, some class contains n.

4. Lower bound

Theorem 4.1. *For all $m \geq 2$, $\ell \geq 2$, and $k \geq n-1$,*

$$S_m(k,\ell) \;\geq\; n-1.$$

Proof. If $n = 1$ the conclusion reads $S_m(k,\ell) \geq 0$, vacuously true. Assume $n \geq 2$. Partition $[1, n-1]$ into singletons $D_r := \{r\}$ for $r = 1, \ldots, n-1$. By Lemma 2.3 each D_r is ℓ-sum-free (since $1 \leq r < n$ forces $n \nmid r$). This is a valid $(n-1)$-partition; for $k \geq n-1$ pad with empty classes by Lemma 2.4. □

Proof of Theorem 1.2. Combine Theorems 3.1 and 4.1. □

5. The threshold $k_0(m,\ell)$

Let $k_0(m,\ell)$ denote the least k such that $S_m(k',\ell) = n-1$ for every $k' \geq k$.

Theorem 5.1. $k_0(m,\ell) \leq \max(1,\, n-1)$.

Proof. Theorem 4.1 exhibits a valid $(n-1)$-partition achieving $N = n-1$ when $n \geq 2$. When $n = 1$ the value is zero, achieved trivially at $k = 1$. □

Theorem 5.2 (Covering lower bound)**.** *Let $\sigma(m,\ell) := \max\{|C| : C \subseteq \{1,\ldots,n-1\},\ C$ is ℓ-sum-free mod $m\}$. Then $k_0(m,\ell) \geq \lceil (n-1)/\sigma(m,\ell) \rceil$.*

Proof. If $[1, n-1]$ admits a valid k-partition, summing class sizes gives $n-1 \leq k \cdot \sigma(m,\ell)$. □

5.1. Tight threshold for prime modulus.

Theorem 5.3 (Large-ℓ coset criterion)**.** *Let $C \subseteq \mathbb{Z}/\mathrm{m}$ with $|C| \geq 2$, pick $a_0 \in C$, and let $H := \langle C - C \rangle = g\,\mathbb{Z}/\mathrm{m}$ with $g := m/|H|$. For every $\ell \geq |H| - 1$,*

$$\ell C \;=\; \ell a_0 + H,$$

and consequently C is ℓ-sum-free modulo m if and only if $g \nmid (\ell-1)a_0$.

Proof sketch. ($\subseteq$) Every ℓ-sum $\sum c_i = \ell a_0 + \sum (c_i - a_0)$ lies in $\ell a_0 + H$ since $c_i - a_0 \in H$.

($\supseteq$) Pick $a_1 \in C$ with $\langle a_1 - a_0 \rangle = H$ (such a_1 exists because differences generate H, and in a cyclic group some single difference generates the full subgroup generated by a finite set). Writing $\delta := a_1 - a_0$, the sequence $0, \delta, 2\delta, \ldots, (|H|-1)\delta$ exhausts H, so for any $h \in H$ there is $j \in \{0, \ldots, |H|-1\}$ with $h = j\delta$; take $c_1 = \cdots = c_j = a_1$ and $c_{j+1} = \cdots = c_\ell = a_0$ (valid since $j \leq |H| - 1 \leq \ell$) to realize $\ell a_0 + h$ as an ℓ-sum.

The safety criterion follows: since $C \subseteq a_0 + H$, we have $\ell C \cap C \neq \emptyset$ iff $\ell a_0 + H = a_0 + H$ iff $(\ell-1)a_0 \in H$ iff $g \mid (\ell-1)a_0$. □

Remark 5.4. The ($\supseteq$) direction is a cyclic-group version of Cauchy–Davenport (cf. Nathanson [**?**], Ch. 2): over $\mathbb{Z}/\mathrm{m}[p]$ prime, ℓ-fold sumsets of any non-degenerate C saturate $\mathbb{Z}/\mathrm{m}[p]$ once $\ell \geq p-1$. Theorem 5.3 is the saturation statement inside an arbitrary cyclic subgroup $H \leq \mathbb{Z}/\mathrm{m}$, with the explicit $\ell \geq |H| - 1$ threshold.

Corollary 5.5 (Sharp k_0 for prime moduli)**.** *Let p be prime. For every $\ell \geq p-1$ with $\ell \not\equiv 1 \pmod{p}$,*

$$k_0(p,\ell) \;=\; p-1.$$

Proof. For any $|C| \geq 2$ in $\mathbb{Z}/\mathrm{m}[p]$, the subgroup $\langle C - C \rangle$ is either 0 (ruled out) or $\mathbb{Z}/\mathrm{m}[p]$ (since $\mathbb{Z}/\mathrm{m}[p]$ has no nontrivial subgroups). By Theorem 5.3 with $g = 1$, $\ell C = \mathbb{Z}/\mathrm{m}[p]$, so C is not ℓ-sum-free. Hence $\sigma(p,\ell) = 1$, and Theorem 5.2 gives $k_0 \geq n-1 = p-1$; Theorem 5.1 supplies the match. □

5.2. **Composite moduli: a counterexample to the natural conjecture.** Corollary 5.4 might suggest: for every m and every large ℓ with $\gcd(m, \ell - 1) < m$, only singletons are ℓ-sum-free, so $k_0(m, \ell) = n - 1$. This is false once m is composite.

Proposition 5.6. *For $m = 12$ and every $\ell \equiv 11 \pmod{12}$ (so $d = 2$, $n = 6$), the set $C = \{1, 5\} \subseteq \{1, \ldots, 5\}$ is ℓ-sum-free modulo 12. Consequently $\sigma(12, \ell) \geq 2$ and $k_0(12, \ell) \leq 3 < n - 1$.*

Proof. $C - C = \{0, \pm 4\}$, so $\langle C - C \rangle = 4\mathbb{Z}/12 = \{0, 4, 8\}$, $|H| = 3$, $g = 4$. Theorem 5.3 applies once $\ell \geq 2$. With $a_0 = 1$, the criterion $4 \nmid (\ell - 1) \cdot 1$ reads $4 \nmid \ell - 1$, i.e. $\ell \not\equiv 1 \pmod 4$. For $\ell \equiv 11 \pmod{12}$ we have $\ell \equiv 3 \pmod 4$, so $4 \nmid \ell - 1$, and C is ℓ-sum-free. Combined with singletons $\{2\}, \{3\}, \{4\}$ this gives a valid 3-partition of $[1, 5]$. □

6. Corollaries: recovered and extended formulas

Specializing Theorem 1.2:

- $m = 8$**:** for $k \geq 7$, $S_8(k, \ell)$ equals $0, 1, 3, 7$ according to $\gcd(8, \ell - 1) \in \{8, 4, 2, 1\}$ (residue pattern $\ell \bmod 8$).
- $m = 9$**:** for $k \geq 8$, $S_9(k, \ell)$ equals $0, 2, 8$ according to $\gcd(9, \ell - 1) \in \{9, 3, 1\}$.
- $m = 11$**:** for $k \geq 10$, $S_{11}(k, \ell) = 0$ if $\ell \equiv 1 \pmod{11}$, else 10.
- **Prime-power** $m = p^e$**:** writing $v := \min(v_p(\ell - 1), e)$, for $k \geq p^{e-v} - 1$,

$$S_{p^e}(k, \ell) \;=\; p^{e-v} - 1.$$

All of these stabilized rows are consistent with (and for $m = 4, 5, 6, 7$ recover) the tables of [3, §2–3].

7. What stays open

(1) *Tight k_0 for composite m.* The covering bound (Theorem 5.2) matches the easy upper bound $n - 1$ for primes but not in general. For $m = 8$, ℓ even: $\sigma(8, \ell) = 4$ (witness $\{1, 3, 5, 7\}$), so the covering bound gives only $k_0 \geq 2$, while the true value is $k_0 = 3$. A closed form for $k_0(m, \ell)$ at composite m is open.

(2) *Boundary $k < n - 1$.* Theorem 1.2 only governs $k \geq n - 1$. The smaller-k regime—where classes must be large and ℓ-sum-free—is empirically eventually periodic in $\ell \bmod m$ for fixed k (verified through $m \leq 13$), but I have no formula.

(3) *Optimizing over ℓ.* For fixed k and $m \to \infty$, what is $\max_\ell S_m(k, \ell)$? The pointwise value is Theorem 1.2; the sup over ℓ becomes a divisor-counting problem on m.

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